THE AI PM BIBLE

Part VI of XIII

Agentic Product Design

Agent-Native UX, Trust Architecture, the Agent Builder Pattern, AI-Native Design Principles, and Measuring What Actually Matters

Jayesh Bachhav
Senior Product Manager · IIM Lucknow · CoEP · CloudBolt | HCL | Kyndryl

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PART VI — AGENTIC PRODUCT DESIGN

Parts I–V built the technical foundation. Part VI is where technology meets the user. It covers the design disciplines unique to AI products — disciplines with no equivalent in traditional software because they deal with probabilistic outputs, multi-step invisible execution, trust as a dynamic variable, and users learning to work with a genuinely new kind of tool. For Jayesh, this is the most directly applicable Part to daily product work — the vocabulary and frameworks for designing products that users actually trust and use.

CHAPTER 30

Agent-Native UX Design — The New Design Vocabulary

Traditional UX is built for deterministic software: click a button, get a predictable result. The entire design vocabulary assumes this model. AI agents violate every assumption: they take time, their actions are not fully predictable, results are probabilistic, they can fail partway through, and they make decisions users need to understand and sometimes override. Designing well for agents requires not adapting old patterns but developing genuinely new ones. This chapter builds the complete design vocabulary for agent-native UX.

30.1 Why Traditional UX Patterns Break for Agents

Traditional UX Assumption	How Agents Violate It	Design Implication
Instant feedback: action → immediate response	Agents take seconds to minutes. The system is actively working.	Progress must be communicated during execution. Silence feels like failure.
Deterministic output: same input → same output	Agents produce probabilistic outputs. Same goal can lead to different execution paths.	Users must understand variation is normal. Trust must survive output variance.
Clear success/failure: worked or didn't	Agent failure is a spectrum: partial completion, low confidence, tool failure.	Failure states must be nuanced. 'Partially completed with caveats' is a valid state.
User always in control	Agents take many actions between user instruction and final result.	Users need visibility into what the agent did, not just what it produced.
Undo is always possible	Some agent actions are irreversible: submitted forms, sent messages, executed process changes.	Reversibility of every action must be communicated before execution.

30.2 The Agent Interaction Lifecycle

Phase	User State	User Need	Design Objective
1. Goal Articulation	Has an objective but may not know how to express it	Help me turn my intent into a well-formed goal	Templates, scope preview, clarifying questions
2. Plan Review	Goal submitted, plan generated	Let me understand what the agent will do before it does it	Human-readable plan; editable; approvable
3. Execution Monitoring	Agent actively executing	Reassure me progress is happening, nothing went wrong	Live progress; interrupt always visible
4. Completion & Results	Agent finished	Help me understand findings and what to do next	Progressive disclosure; confidence signals; next actions
5. Reflection & Learning	User acted on results	Let the system improve from my feedback	Lightweight feedback; episodic memory

30.3 Goal Setting and Task Initiation UX

The most underdesigned phase of agent UX is goal articulation. Most agent products present users with a blank text box and expect them to know how to formulate a precise, well-scoped goal. This is a design failure, not a user failure.

Provide goal templates for common task types

Most users ask a small number of predictable question types. Provide pre-built templates: 'Investigate anomaly: [parameter] at [plant] on [date]', 'Generate performance report: [plant] for [time period]'. Templates reduce cognitive load and produce better-scoped goals than free-form input.

Show scope preview before execution

Before the user submits a goal, show them: what data will be accessed, what tools will be used, approximately how long execution will take, and what the output will look like. This catches misunderstandings before they become wasted agent runs.

Ask 1-2 clarifying questions for ambiguous goals

Never start executing an ambiguous goal and fail halfway through. The cost of a clarification question is seconds. The cost of an aborted execution is minutes and lost trust.

Show recent successful goals as examples

Display 3-5 recently successful goals from the same user or similar users. Teaches how to formulate effective goals through example without documentation.

30.4 Transparency During Execution

Communication Level	What to Show	When to Use
Level 0 — Spinner Only	Progress animation. No detail.	Simple, fast tasks < 10 seconds.
Level 1 — Status Message	Brief text: 'Retrieving sensor data...', 'Generating analysis...'	Tasks 10–60 seconds. Reassures without technical detail.
Level 2 — Step Progress	Named steps with completion status: checkmarks, spinner, grayed-out pending.	Tasks 1–5 minutes. Shows how far along and how much longer.
Level 3 — Live Action Feed	Real-time stream: 'Querying historian for turbidity 14:00–16:00...', 'Found 3 similar incidents...'	Tasks > 5 minutes or high-stakes tasks requiring full visibility.
Level 4 — Reasoning Stream	Token-by-token streaming of the agent's thinking before presenting results.	Power users and debugging; not the default.

The Always-Visible Interrupt Control

Regardless of execution communication level, one control must ALWAYS be visible: the interrupt button. Design requirements: always visible during execution, never hidden by progress animations; clearly labeled 'Stop' or 'Pause & Review'; on interrupt, immediately pause, save state, show partial results so far; on interrupt, offer 'Resume', 'Restart with changes', or 'Cancel'. Most users never press it — but its presence reduces anxiety during long-running tasks and communicates: you are always in control.

30.5 Progressive Disclosure of Agent Reasoning

Disclosure Level	Content	Who Sees It
Level 0 — Result Only	The final answer or recommendation.	All users, by default
Level 1 — Key Evidence	Top 2-3 evidence items. Source documents cited.	Users who click 'Why?' or 'Show evidence'
Level 2 — Reasoning Summary	Paragraph explaining how the agent reached its conclusion.	Users who want to validate reasoning before acting
Level 3 — Step-by-Step Trace	Complete execution trace: every step, tool call, result.	Engineers debugging; auditors; power users
Level 4 — Raw Prompt/Completion	The raw LLM input and output.	Developers only, via observability tooling — never in product UI

Progressive Disclosure — Worked Example

DEFAULT (Level 0): Recommendation: Increase PAC dose from 2.2 to 2.6 mg/L [Why?] [Sources] [Approve] [Reject]. EXPANDED (Level 1 — after 'Why?'): (1) Turbidity trending up 18% over 2 hours. (2) pH stable at 7.3 — optimal coagulation range. (3) This adjustment resolved similar conditions in most comparable historical cases. EXPANDED FURTHER (Level 2 — 'Full reasoning'): Agent analyzed recent process data, identified an upward turbidity trend not explained by flow rate changes, and compared it against historical incidents. The PAC dose adjustment was the most common successful intervention, and the recommendation is within the normal operating range.

30.6 The Approval and Intervention Interface

Element	Purpose	Design Requirements
Recommendation Display	Present the agent's recommendation clearly	Lead with the action. Rationale follows. Risk level shown visually.
Evidence Panel	Allow inspection before deciding	Expandable, not default focus. Linked to source documents. Confidence per item.
Modification Interface	Allow user to adjust before approving	Inline editing. Agent recommendation alongside user modification. Preview the modified action.
Approve / Reject / Defer	Capture decision with minimal friction	Approve is most prominent. Reject and Defer available. Rejection offers one-tap reason.

Element	Purpose	Design Requirements
Consequence Preview	Show what approving will do	Irreversible: state it clearly. Reversible: show how to undo.
Audit Record	Capture full decision context	Automatic. Includes what agent recommended, what user decided, and any modification.

30.7 Completion and Result Presentation

Once an agent finishes, the job shifts from reassurance to orientation: the user needs to understand what happened and what to do next, fast — not re-read the whole execution trace.

Element	What to Show	Design Guidance
Lead Summary	The direct answer or outcome, not a list of steps taken.	State the conclusion first. Evidence and process are available, not default.
Confidence	How reliable this specific result is.	Pair with supporting evidence, never a bare number alone.
Comparison to Expectation	How this result compares to a typical or prior result.	Helps the user calibrate whether to dig deeper or move on.
Next Actions	What the user can do right now.	Offer 2-3 concrete options rather than an open-ended prompt.

30.8 Error and Failure UX for Agents

Failure Type	User-Facing Message Pattern	Recovery Path
Complete failure	'I wasn't able to complete this. [Reason]. [What I tried before stopping].'	Retry with different parameters; route to human expert
Partial completion	'I completed N of M steps. Here's what I found: [partial]. I couldn't complete [remaining] because [reason].'	Accept partial results; resume; try alternative
Low confidence result	'I completed the analysis but have low confidence. [Results]. Recommend verifying with [resource] before acting.'	View evidence; request human review; try alternative sources
Information not found	'I searched [sources] for [topic] and found no relevant documentation. This may not be in our knowledge base.'	Suggest adding docs; route to expert; web search fallback
Out of scope	'This is outside my knowledge area. I can help with [specific scope]. For this, I suggest [resource].'	Clarify scope; route to appropriate resource

30.9 The Agent UX Design Checklist

Agent UX Design Checklist — Before You Ship

GOAL ARTICULATION: templates for the most common task types; scope preview before execution; clarifying questions for ambiguous goals; example goals for new users. EXECUTION: progress communication appropriate to task length; interrupt control always visible; estimated time for tasks over 30 seconds; partial results preserved if user interrupts. RESULTS: progressive disclosure — result first, evidence on demand, trace for experts; confidence indicators on all outputs; source citations for all factual claims; clear next actions presented. APPROVAL: consequence preview for irreversible actions; approve/modify/reject for all recommendations; audit record automatically captured. FAILURE: specific non-generic message for each failure type; recovery path offered for every failure; partial results preserved and shown when a task fails mid-execution.

CHAPTER 31

Trust Architecture — Designing AI Products That Earn and Maintain Trust

Trust is not a feature you add — it is an architecture. It is the accumulated result of hundreds of design decisions, each of which either builds or erodes confidence. The AI products that achieve deep adoption are not the most capable ones — they are the ones that have built the deepest trust. For industrial AI where errors have real-world consequences, trust architecture is the central product design challenge.

31.1 Trust as Architecture, Not a Feature

The most common mistake in AI product design is treating trust as a feature: 'we'll add a confidence score.' This is useful but it is a component of trust, not trust itself. Trust emerges from the totality of the user's experience: how the product behaves when right, when wrong, when uncertain, when it fails, when challenged.

> **KEY INSIGHT**
>
> Trust in AI is asymmetric: it takes many positive interactions to build and a single confident hallucination in a high-stakes moment to destroy. Focus more design effort on failure modes than on the happy path. The happy path builds trust incrementally. How the product handles failures determines whether trust is durable.

31.2 The Trust-Building Sequence

Trust Stage	User Mental Model	Design Objective	Key Metric
Skeptical Exploration (Sessions 1–3)	'Let me see if this is actually good.' Testing on questions they know the answer to.	Demonstrate competence on most common tasks. Never fail on easy questions.	Retention to session 3. User completes 3 full interactions.
Conditional Use (Sessions 4–20)	'It's good for X but I don't trust it for Y.' Selective use where AI has proven reliable.	Reinforce reliability on proven task types. Extend carefully into new types.	Breadth of task types attempted.
Functional Trust (Sessions 20–100)	'I rely on it for specific tasks.' Regular use with appropriate verification.	Enable deeper reliance. Surface new capabilities at moments of curiosity. Build episodic memory.	Daily active use. Recommendation follow-through rate.
Expert Partnership (100+ sessions)	'I know what it's good at.' High productivity, calibrated trust.	Power user features: deeper reasoning access, customization, feedback loops.	Time saved per task. NPS > 50.

31.3 Calibrated Trust — The Target State

The goal is not maximum trust — it is calibrated trust. A user with calibrated trust knows when to follow the AI's recommendations directly and when to verify before acting. This is more valuable than maximum trust because it produces better outcomes and is more durable.

WATCH OUT: Automation Bias — The Most Dangerous Trust Failure

Automation bias is the well-documented tendency to over-rely on automated systems — accepting outputs without critical evaluation, especially under time pressure. In industrial settings, automation bias is a safety risk. An operator who follows an AI dosing recommendation without checking the supporting data is exhibiting automation bias. Product design must actively counteract this: surface confidence indicators, require explicit confirmation for safety-adjacent actions, and periodically require users to engage with evidence before approving. Do not design for maximum compliance — design for calibrated judgment.

31.4 Trust Signals — What to Show and When

Trust Signal	What It Communicates	Design Implementation
Source Citations	This answer is grounded in specific documents I can verify	Inline citation. One-click to view source. Always shown on factual claims.
Confidence Indicator	How confident the system is in this specific answer	Color-coded (green/yellow/orange) with text label. Never numerical percentages — users miscalibrate them.
Uncertainty Acknowledgment	The system knows the limits of its knowledge	Plain language: 'Based on available data', 'I don't have information on this'.
Data Freshness	When was this information last updated	Timestamp on documents. Warning when > 6 months old. Signal when real-time data unavailable.
Scope Indicator	This answer is within or outside the system's reliable domain	In-scope: no indicator. Out-of-scope: 'This is outside my primary area — verify with [resource]'.
Track Record	Historical accuracy on similar tasks	Aggregate accuracy shown in onboarding. Per-task-type accuracy available on demand.

31.5 Trust Recovery After Failure

Failure Type	Wrong Response	Right Response
Hallucination discovered	Double down or ignore feedback	Immediate clear acknowledgment: 'You're right — that was incorrect. I apologize.'
Task failed mid-execution	Generic error, no context	'I attempted [steps]. Failed at [step] because [reason]. Here's what I found before failing.'
Recommendation led to poor outcome	No acknowledgment	'I understand this didn't work out. Can you tell me what happened? This helps me improve.'
User repeatedly rejects recommendations	Continue generating similar ones	Detect pattern: 'I've noticed you've adjusted my last 3 recommendations. Shall I account for this going forward?'

31.6 The Confidence Communication Patterns Library

The Soft Hedge Pattern

Language hedges embedded naturally signal uncertainty without explicit labels. 'Based on available sensor data, the most likely cause is...' vs. 'The cause is...' The hedge is semantic content that helps calibration. Requires only prompt engineering — no UI change.

The Confidence Badge Pattern

A small visual badge indicates confidence: green for high, yellow for moderate, orange for low. Must be explained in onboarding. The fastest way to scan multiple recommendations. Risk: color alone is inaccessible — always pair with text label.

The Evidence-First Pattern

Lead with the most important evidence and derive the conclusion from it. Users trust conclusions they can trace back to evidence more than conclusions with post-hoc support.

The Explicit Abstain Pattern

AI declines to answer when confidence is below threshold. 'I don't have sufficient data for a reliable recommendation. I recommend reviewing with your process engineer.' This is deeply trust-building — it demonstrates the system knows what it doesn't know.

The Track Record Display Pattern

Show historical accuracy on the same task type. 'This type of anomaly analysis has been accurate in the large majority of similar past cases.' The most credible confidence signal — but requires tracking actual outcomes.

31.7 Trust Research Methods for AI Products

Trust is a claim worth testing, not assuming. These are the methods that actually reveal whether users trust a system appropriately — not just whether they say they do.

Method	What It Reveals	When to Use
Calibration testing	Whether users' stated trust matches their actual behavior — do they verify when they should?	Before and after major model or UX changes
Override rate analysis	Where users are overriding the AI, and why	Ongoing, as a leading indicator
Think-aloud sessions	How users reason about AI outputs in real time	Early design validation
Trust incident review	What specific moments eroded trust and why	After any trust-damaging failure
Longitudinal cohort tracking	How trust changes across the trust-building sequence (Section 31.2) over weeks or months	Ongoing product health monitoring

31.8 Industrial Trust Design Framework

Industrial Trust Design Framework — Six Non-Negotiable Principles

PRINCIPLE 1 — NEVER HIDE UNCERTAINTY: Every operational recommendation must include a confidence indicator and data basis. No recommendation may be presented without its evidentiary foundation. PRINCIPLE 2 — CALIBRATION BEFORE RELIANCE: New operators complete a mandatory calibration experience: verified historical scenarios where the AI is correct, and scenarios where it is suboptimal. Builds an accurate mental model before operational use. PRINCIPLE 3 — GRADUATED AUTONOMY: Start at full oversight. Graduate to reduced oversight only after an operator demonstrates consistent appropriate judgment across many interactions. PRINCIPLE 4 — FAILURE TRANSPARENCY: When wrong, acknowledge it clearly and quickly. Low-friction error reporting. Show operators their history, including cases where they correctly overrode the AI. PRINCIPLE 5 — HUMAN ALWAYS WINS: Operator's decision always overrides AI with no friction. AI may ask once 'Are you sure?' but must execute the operator's decision. Never require justification for overriding. PRINCIPLE 6 — AUDIT VISIBILITY: Every operator can see their complete interaction history: what AI recommended, what they decided, and what happened. Transparency builds confidence in the system as a fair partner.

CHAPTER 32

The Agent Builder Pattern — Configurable Agents for Non-Technical Users

The Agent Builder pattern solves a fundamental enterprise AI problem: the organizations that most need AI agents often lack the engineering resources to build them. Agent Builder provides a no-code or low-code interface that lets non-technical users configure, deploy, and manage AI agents tailored to their specific workflows. One real-world deployment of this pattern gives plant managers and process engineers the ability to create custom analysis and automation agents without writing code.

32.1 What Is the Agent Builder Pattern?

Agent Builder Layer	What the User Configures	What the Platform Handles
Identity & Purpose	Agent name, description, role in plain language	Translation to system prompt instructions
Knowledge Sources	Which documents, databases, data sources the agent can access	RAG corpus scoping, embedding, retrieval pipeline
Tools & Actions	What the agent can do: query systems, generate reports, send notifications	Tool schema generation, API integration, permission enforcement
Interaction Model	How users interact: chat, workflow trigger, scheduled, event-driven	Conversation management, trigger handling, scheduling
Governance	Who can use it, approval requirements	Permission enforcement, guardrails, audit logging

32.2 The Abstraction Layers — What to Expose vs. Handle in Platform

Decision	Expose to User	Handle in Platform	Rationale
Agent's purpose and domain	Yes — plain language	No	Core to the agent's value.
Which knowledge sources are available	Yes — checkbox list	No	User knows their data best.
Which tools the agent can use	Yes — toggle list	No	User decides what agent is allowed to do.
System prompt structure	No	Yes — derived from description	Too technical. Platform translates intent.
Embedding model selection	No	Yes — platform chooses	Too technical. Platform knows best performer.
Temperature and parameters	No by default; advanced mode option	Yes for standard mode	Most users don't know what temperature does.
Chunking strategy	No	Yes — by document type	Too technical.

Decision	Expose to User	Handle in Platform	Rationale
Guardrail configuration	Partial — risk tolerance, prohibited topics	Platform enforces minimum guardrails always	Users can customize tone, not disable safety.

32.3 The Agent Configuration Interface — Six Steps

1. Name and Purpose (Plain Language)

Ask: 'What is this agent's job?' Name field (e.g. 'Plant 3 Shift Handover Assistant'), description, optional avatar. Platform uses this to generate the system prompt automatically. Show generated prompt in collapsible 'Advanced' section for technical users.

2. Knowledge Sources

List of available source types: company documentation (file upload or SharePoint/Google Drive), databases (historian, incidents, maintenance), external sources. Each source has scope: which plants, time ranges, equipment. Checkbox + scope selector.

3. Tools and Capabilities

Toggle list: read sensor data, search incident history, generate PDF report, send email, create work order, query regulatory database. Each toggle has an info tooltip. No capability enabled by default — users must explicitly grant.

4. Interaction Model

How do users interact? Options: chat interface, triggered workflow (event-driven), scheduled report, on-demand analysis. Select one. Determines deployment template.

5. Test Before Publish

Creator tests with three scenarios (easy, moderate, edge case) before the agent is available to others. Platform shows quality metrics: right knowledge sources used? Stayed in scope? Output format correct? Creator confirms before publishing.

6. Governance Configuration

Who can use it? (Specific users, roles, or all staff.) Approval requirements? Notification settings. These settings determine the trust and oversight model for the deployed agent.

32.4 Template Agents vs. Custom Agents

Template Category	Example Agents	Customization Surface
Process Analysis	Turbidity Spike Investigator, PAC Dosing Optimizer, Chlorination Compliance Checker	Plant scope, time window, threshold parameters, output format

Template Category	Example Agents	Customization Surface
Shift Operations	Shift Handover Brief Generator, Morning Status Report, Active Alarm Summarizer	Plant scope, included parameters, distribution list, format
Maintenance	Equipment Fault Diagnostician, Preventive Maintenance Scheduler, Spare Parts Checker	Equipment scope, maintenance schedule source, work order integration
Compliance & Reporting	Monthly Regulatory Report Generator, DPR Automation, Water Quality Trend Report	Regulatory parameters, reporting period, format requirements
Training & Knowledge	SOP Lookup Agent, New Operator Guide, Chemical Safety Advisor	Document scope, operator role, language preferences

32.5 The Agent Marketplace

Once an organization has more than a handful of agents, discovery becomes the bottleneck — not creation. A marketplace layer turns individual agent creation into platform-wide leverage.

Capability	Purpose	Design Consideration
Publish	Let agent creators share a working agent as a reusable template	Require testing before publish; show quality signals, not just a submit button
Discover	Help users find a relevant existing agent instead of building from scratch	Search and categorize by use case, not only by creator
Fork	Let a user start from an existing agent and customize it	Preserve attribution to the original creator
Rate and review	Surface which agents are actually reliable	Weight by real usage outcomes, not star ratings alone

32.6 Permissioning and Governance

The Four-Layer Governance Architecture

Layer	Who Controls It	What It Governs
Platform Layer (platform-defined)	Platform team	Absolute limits: no process control actions, mandatory audit logging, mandatory guardrails
Organization Layer (customer admin)	Customer IT/security admin	Which data sources available, which tools can be enabled, which roles can build agents
Agent Layer (creator-defined)	Operator or engineer who built the agent	Scope within enabled options: specific sources, specific tools, interaction model
User Layer (end user)	Operator using the agent	Cannot exceed agent creator's permissions. Personal preferences within settings.

32.7 Example Agent Builder — Product Specification

Example Agent Builder — Complete Product Specification

VISION: Any customer — plant manager, process engineer, shift supervisor — can configure a custom AI agent for their specific workflow in under 30 minutes, without code, with full governance and safety controls. CORE USER STORIES: 1. Plant Manager: create a shift handover agent that summarizes key events automatically. 2. Process Engineer: build an anomaly investigation agent that queries sensor data when a turbidity alarm fires. 3. Operations Director: deploy agents from templates and customize for plant-specific parameters. V1 MINIMUM VIABLE FEATURE SET: agent creation wizard (six-step flow as specified above); starter templates covering core use cases; knowledge connectors (process historian, incident history, document library); tool catalog (query_historian, search_incidents, search_docs, generate_report); chat interaction model; test-before-publish with quality indicators; role-based access control for agent deployment; audit log of all agent interactions. V2 ADDITIONS: triggered workflow and scheduled report models; agent marketplace (publish and discover); advanced configuration mode for technical users; agent performance dashboard per agent. NON-NEGOTIABLE GUARDRAILS (cannot be disabled by any user): no process control system integration; mandatory safety confidence thresholding; mandatory audit logging with long-term retention; mandatory human approval for elevated-risk recommendations.

CHAPTER 33

AI-Native Design Patterns — Beyond Adapting Old Patterns

Most AI product UX today is adapted traditional UX: add AI results to a search box, add AI autofill to a form, add AI insights to a dashboard. These work as a starting point but do not unlock AI's transformative potential. Truly AI-native design starts from what AI makes possible rather than from what traditional software has done. This chapter catalogs nine patterns that are native to AI — patterns that could not exist without the probabilistic, generative, context-aware capabilities of modern language models.

33.1 The Paradigm Shift — Task Execution to Goal Pursuit

Dimension	Task Execution Design	Goal Pursuit Design
User input	Specify the exact action: 'Sort by date'	Express the intent: 'Show me the most urgent items'
System response	Execute the specified action deterministically	Interpret intent, determine approach, execute, explain
Error handling	'Invalid input' — user must fix specification	'I interpreted this as X — is that right?' Collaborative clarification
Workflow	User defines the steps	User defines the outcome; system proposes the path
Interface primitives	Buttons, forms, dropdowns	Natural language, approval flows, progressive disclosure

33.2 The Ghost Text Pattern

Ghost text renders AI suggestions inline with lighter gray styling ahead of the cursor. Accepted by pressing Tab; ignored by continuing to type. The cognitive load is minimal: one keystroke accepts, no action ignores. The design pattern behind GitHub Copilot's tens of millions of daily users and Notion AI. For example: suggest completions in maintenance log entries, recommend parameter values in report fields, complete shift notes based on sensor data.

33.3 The Ambient Intelligence Pattern

Traditional AI requires the user to go to the AI. Ambient intelligence brings the AI to the user — surfacing insights, alerts, and recommendations proactively in the user's natural workflow, without an explicit query. Requires careful calibration: too many suggestions become noise that users learn to ignore.

Ambient Intelligence — Worked Example

SIGNAL 1 — Proactive Alert (high urgency): an amber indicator appears on a trend chart without operator action. On hover: 'Trending up over the last 90 minutes. Similar patterns preceded upsets in several historical matches. Consider checking the relevant control setting.' [View Analysis] [Dismiss] [Remind me in 30 min]. SIGNAL 2 — Contextual Suggestion (medium priority): as the operator prepares a handover report, the assistant sidebar shows: 'Based on today's data, a few items worth including: [1] a trend worth noting [2] equipment maintenance logged [3] a flow rate change' [Add All] [Add Selected] [Dismiss]. SIGNAL 3 — Learning Suggestion (low urgency): 'You've run a similar analysis several times this week. Would you like to set up a daily automated version?' [Set Up Agent] [Not Now] [Don't Ask Again]

33.4 The Collaborative Refinement Pattern

Traditional software: user accepts or rejects output. AI enables collaborative refinement — an iterative process where user and AI improve an output together. AI produces a draft; user annotates, corrects, or redirects; AI revises; user refines. For example: a supervisor uses AI to draft a root cause analysis, highlights disagreements, the AI produces a revised draft incorporating the corrections, and the supervisor approves. The final report is better than either would produce alone.

33.5 The Proactive Surface Pattern

Context Signal	Proactive Content	Implementation
User opens app at shift start	Shift briefing: key events, active alarms, items requiring attention	Event-triggered agent at shift transition
Sensor reading crosses threshold	Alert with context and suggested response	Threshold monitor → anomaly agent → notification
Regulatory reporting deadline approaching	Data completeness status: what's ready, what's missing	Calendar-triggered reminders before deadline
New documentation in knowledge base	Summary of newly available information relevant to user's role	Corpus ingestion trigger → role-relevant summary

33.6 The Confidence Gradient Pattern

Rather than binary trust signals, content is rendered with visual styling that varies continuously with confidence. High-confidence content: full saturation. Lower-confidence: progressively lighter, muted, or italicized. Users immediately identify which portions warrant careful attention. The traditional disclaimer at the bottom is almost never read — the confidence gradient is unavoidable as the user reads.

33.7 The Reversible Action Pattern

Action Type	Reversibility	UX Treatment
Fully reversible	Can be undone immediately	Low friction. Single-click approve. Undo always visible after.
Reversible with effort	Undoing requires action	Medium friction. Confirm first. Clear undo path after.

Action Type	Reversibility	UX Treatment
Difficult to reverse	Undoing requires significant effort	High friction. Explicit confirmation with consequence stated.
Irreversible	Cannot be undone	Highest friction. Double confirmation. Supervisor approval. Never automated.

33.8 The Explanation on Demand Pattern

A 'Why?' button always present and always returning a genuine, specific explanation — not a generic statement about how AI works. 'I recommended increasing the dose because turbidity has risen 18% and similar historical incidents were resolved with this adjustment' is a real explanation. 'This recommendation was generated by an AI analyzing your process data' is not. The explanation must be specific to this output, this data, this query.

33.9 The AI-Native Onboarding Pattern

AI-Native Onboarding — Worked Example

STEP 1 — COMPETENCE DEMONSTRATION (5 min): show pre-run examples where the assistant answered correctly with verifiable citations, relevant to the user's own context. Goal: 'This could be helpful.' STEP 2 — LIMITATION ACKNOWLEDGMENT (3 min): show pre-run examples where the assistant said 'I don't know' appropriately. Explain: this is the system working correctly. Goal: 'I can trust it when confident, verify when uncertain.' STEP 3 — INTERACTIVE TRIAL (10 min): user asks several real questions relevant to their own work. Coach on confidence indicators and citations. Let them try overriding a recommendation. Goal: muscle memory for interaction patterns. STEP 4 — SCOPE CALIBRATION (2 min): tell the user what the assistant knows well and what it doesn't. Goal: accurate scope mental model. STEP 5 — FEEDBACK INTRODUCTION (1 min): show the feedback button. Explain corrections improve future recommendations. Ask the user to flag at least one response in their first week.

CHAPTER 34

Agent Product Metrics — Measuring What Actually Matters

Traditional SaaS metrics were built for deterministic software with clear usage signals. Agent products need a different measurement model — one that captures quality, trust, and value, not just activity. Agent products also introduce a cost-efficiency dimension that makes them commercially viable. This chapter builds the complete metrics framework — from the foundational task completion rate to the nuanced autonomy ratio that measures maturity.

34.1 Why Traditional SaaS Metrics Fail for Agent Products

Traditional Metric	Why It Fails	Better Alternative
Daily Active Users (DAU)	High DAU may reflect novelty-driven exploration, not value delivery.	Recommendation follow-through rate: what fraction of agent recommendations do users act on?
Session Length	Longer sessions may mean the agent is too slow or the user is re-running failed tasks.	Tasks completed per session. Time saved vs. manual baseline.
Feature Adoption Rate	A user who tries a feature once shows 100% 'adoption.' Meaningless for agents.	Recurring agent use: fraction using the same agent repeatedly within 30 days.
Churn Rate	Enterprise contracts mask churn. Customer can be paying and not using.	Value realization rate: is the customer achieving their stated use case?
NPS	Measures satisfaction, not value delivery.	Autonomy ratio: fraction of interactions requiring no human correction.

34.2 The Four-Layer Agent Metrics Framework

Layer	Category	What It Measures	Leading or Lagging
L1	Quality	Task completion, faithfulness, accuracy — is the agent doing its job well?	Leading — must be high before adoption grows
L2	Trust	Follow-through rate, autonomy ratio, override rate — are users trusting and acting on outputs?	Leading — trust precedes retention
L3	Value	Time saved, decision quality, error rate reduction — measurable business value?	Lagging — outcome of quality + trust
L4	Scale	User growth, agent diversity, tasks per day — growing and expanding?	Lagging — result of quality + trust + value

34.3 Task Completion Rate — The Foundation Metric

Sub-Metric	Definition	Target	Alert Threshold
Full completion rate	Agent completes all steps and produces intended output	> 88%	< 80%

Sub-Metric	Definition	Target	Alert Threshold
Partial completion rate	Completes some steps, produces useful partial output	Acceptable < 10%	If > 15%, investigate step failure patterns
Refusal rate	Agent declines to attempt the task	Acceptable 2–5%	If > 8%, prompt regression or scope issue
Error rate	Agent encounters unhandled error	< 1%	If > 2%, engineering investigation required
Timeout rate	Agent exceeds time limit	< 2%	If > 5%, performance or complexity issue

34.4 Trust Velocity — The AI-Specific Retention Signal

Trust velocity measures how quickly a new user progresses through the trust stages. It is the single most predictive metric for long-term retention — more predictive than DAU, session length, or task completion rate. Measured as rate of change in recommendation follow-through rate (increasing over time) and override rate (decreasing over time). Flat or declining follow-through means trust is not building; investigate quality and UX.

34.5 Autonomy Ratio — The Maturity Metric

Autonomy ratio: fraction of agent interactions that complete without human correction, modification, or override. It is the primary measure of agent product maturity.

Autonomy Ratio	Interpretation	Stage
< 40%	Agent is unreliable or trust not yet built. Most interactions require correction.	Early pilot. Focus on quality improvements and trust-building UX.
40–65%	Agent delivers value but users actively supervising most interactions.	Growing trust. Focus on reducing systematic failure patterns.
65–80%	Agent trusted for routine tasks. Users verify on novel or high-stakes interactions.	Operational maturity. Focus on expanding task type coverage.
80–90%	High trust for deployed task types. Oversight reserved for edge cases.	High maturity. Focus on expanding scope and agent portfolio.
> 90%	Only appropriate for most routine, fully validated task types.	Fully mature. Review whether adequate oversight is still occurring.

34.6 Time-to-Value per Task

Task Type	Manual Baseline	AI-Assisted Target	ROI
Turbidity spike root cause analysis	2–4 hours	15–30 minutes	4–8x time saving
Shift handover report	30–60 minutes	5–10 minutes	4–6x time saving

Task Type	Manual Baseline	AI-Assisted Target	ROI
Monthly regulatory report	2–3 days	4–6 hours	4–6x time saving
Equipment fault diagnosis	1–3 hours	20–40 minutes	3–4x time saving

34.7 The Agent Product Health Dashboard

> **Weekly Agent Health Review Metrics**
>
> QUALITY: task completion rate (> 88%); RAGAS faithfulness (> 0.87); refusal rate (2–5%); error reports per 1000 queries (< 3). TRUST: recommendation follow-through rate (increasing); autonomy ratio (moving toward 65–80%); override rate (decreasing). VALUE: average time-to-value per task (< 30 min for core tasks); tasks completed vs. manual baseline. OPERATIONAL: P95 latency (< 8s); error rate (< 1%); daily cost per active user (decreasing). GROWTH: weekly active users; new agent types configured (Agent Builder); template vs. custom ratio.

34.8 Metrics Anti-Patterns

Anti-Pattern	Why It's Dangerous	What to Do Instead
Optimizing approval rate without checking outcome quality	High approval may reflect automation bias, not AI quality.	Track outcomes: when users approve recommendations, do they actually work?
Using general benchmark scores as quality proxies	Benchmarks don't predict performance on your specific tasks.	Build and track product-specific quality metrics.
Setting autonomy ratio as pure north-star	Chasing high autonomy without monitoring quality leads to removing oversight, increasing risk.	Track autonomy alongside outcome quality. Autonomy is good only when outcomes are good.
Ignoring override rate as just negative engagement	High override rate is an AI quality failure signal, not user engagement.	Investigate what recommendation types are most commonly overridden.

CHAPTER 35

Operator Experience Design — AI for the Industrial Workforce

Most AI product design is created for knowledge workers in office environments: reliable internet, large screens, time for deliberation, high AI literacy. Industrial operators are a fundamentally different user population. They work in high-cognitive-load environments, are under time pressure, may have limited AI literacy, and use mobile devices or rugged tablets in the field. The consequences of an unclear recommendation or confusing interface are not a wasted work hour — they can be a process upset or a safety incident.

35.1 The Industrial Operator as AI Product User

Characteristic	Design Implication
Deep domain expertise, variable AI literacy	Design for domain experts, not AI experts. Use domain vocabulary. 'Based on sensor data' not 'retrieved from the RAG corpus.'
High cognitive load during active operations	UI must be minimal and decisive during operations. Save complexity for analytical tasks performed between events.
Time pressure during incidents	During an alarm, operator needs the recommendation in < 5 seconds, not a 3-paragraph analysis. Context-aware mode switching.
12-hour shift rotations with handover	Continuity across shifts is critical. AI should help with handover and preserve context between shifts.
Physical environment: noise, PPE, glare	Large touch targets (gloves), high contrast (glare), no audio-critical features unless headset confirmed.
Accountability culture	Operators are accountable for their decisions. AI must be assistant, never replacement for judgment. Override with no friction.
Varying languages	In India, operators may communicate in Hindi or regional languages. English-only AI creates barriers.

35.2 Cognitive Load in High-Stakes Environments

Cognitive Load Type	Industrial AI Design Implication
Intrinsic — managing an active process event	Cannot be reduced — this is the operator's job. Design must not add to it. AI during active events should be a single-action recommendation, not an analysis workflow.
Extraneous — understanding the AI interface	Minimize: consistent, predictable patterns. No new vocabulary without explanation. No unnecessary choices during high-load moments.
Germane — learning to work effectively with AI	Preserve: operator should develop skill over time. Calibration experiences and gradual autonomy expansion build germane understanding.

35.3 Designing for the Shift Change Context

Shift Change Need	AI Design Solution	Implementation
Complete picture of current state	AI-generated shift status snapshot: all parameters, alarms, pending actions	Scheduled agent 30 minutes before shift end. Queries all live data.
Significant events from last shift	Prioritized event timeline with impact assessment	Agent analyses events, ranks by operational significance, generates narrative.
Ongoing investigations	Context preservation across shift boundary	Persistent agent memory: active investigations automatically surfaced to incoming shift.
Items requiring follow-up	Actionable handover items with ownership	Agent generates checklist. Outgoing reviews. Incoming acknowledges each item.

35.4 Mobile-First Industrial AI Design

Requirement	Specification
Touch target size	Minimum 44x44px standard; 48x48px for gloved operation; critical actions (Approve, Stop) 60px+
High contrast	WCAG AA minimum. Outdoor/bright light: high-contrast mode with 7:1 contrast ratio.
One-handed operation	Primary actions reachable with thumb. Bottom navigation. Critical decisions accessible one-handed.
Offline resilience	Core functionality (view state, view alerts, view SOPs) works offline. Queue actions, sync when connected.
Response time	P95 < 3 seconds on 4G. P95 < 6 seconds on 3G. Skeleton loading states, never blank screens.
Session persistence	Operator leaves app, returns 30 minutes later. Session exactly where they left it. No re-authentication mid-shift.

35.5 Multilingual Operator Experience

Component	Implementation Approach
UI language localization	Full localization to Hindi + regional languages. Not just labels — all error messages, tooltips, onboarding content.
Query processing	Accept Hindi or mixed Hindi-English (Hinglish). Multilingual embedding models handle mixed-language queries.
Response generation	Generate in the user's preferred language. Modern LLMs generate high-quality Hindi. Specify language in the system prompt.
Voice input	Speech-to-text support for Hindi + regional languages. Reduces text input barrier significantly.
Domain terminology	Technical terms consistently translated or transliterated. Domain glossary included in system prompt.

35.6 Designing for Varying AI Literacy Levels

Even within one plant, operators range from AI-skeptical veterans to AI-native new hires. Treating them as one audience under-serves both.

Literacy Level	What This User Needs	Design Implication
New to AI tools	Reassurance the system won't act without them	Conservative defaults. Explicit confirmation before any action.
Comfortable with AI, new to this product	A fast path to understanding what this agent specifically can and cannot do	Short, concrete onboarding (Section 33.9) over generic AI explainers.
Experienced power user	Efficiency, not hand-holding	Shortcuts, advanced mode, reduced confirmation friction once trust is calibrated.

35.7 Operator Experience Design Specification — Worked Example

Copilot — Operator Experience Specification (Worked Example)

CONTEXT MODES (auto-switching based on detection): ACTIVE MODE (alarm firing, process upset detected): single-action recommendation card. One approve/reject. No prose. High contrast. Response target: under 5 seconds. Max 50 words. ROUTINE MODE (normal operations): standard chat. Full recommendations with evidence. Response target: under 10 seconds. ANALYSIS MODE (investigation or report requested): full agent execution with progress visibility. Extended times acceptable. Response target: under 5 minutes for full investigation. MOBILE REQUIREMENTS: touch targets 48px minimum; high contrast enabled by default; voice input in Hindi + English on the main query field; offline mode with the last 24 hours of data and the SOP library cached locally; session persistence of a full 12-hour shift with no re-authentication. LANGUAGE SETTINGS: default set during onboarding; English and Hindi available (regional languages in a later version); mixed-language queries accepted without explicit language selection. SHIFT CHANGE: automated shift snapshot 30 minutes before shift end; handover checklist agent-created, supervisor-reviewed; incoming shift briefing context-aware on first login after shift change. TRUST UX: confidence indicator always visible on recommendations; source citation one-tap from every recommendation; override always available, no friction, no justification required; calibration tour mandatory for all new operators; outcome feedback optional one-tap after every approved recommendation.

PART VI — SYNTHESIS AND KEY TAKEAWAYS

What You Now Know — And Why It Matters

Part VI has covered the design layer of agentic AI products — the discipline that determines whether technically capable AI systems are actually used, trusted, and valuable. Agent-native UX, trust architecture, the Agent Builder pattern, AI-native design patterns, agent metrics, and industrial operator experience: these are the frameworks that separate AI products that change how people work from AI features that are tried once and forgotten.

Core Insight	Why It Matters	PM Action
Agent UX requires a new design vocabulary: goal articulation, execution transparency, progressive disclosure, approval interfaces, nuanced failure states.	Traditional UX patterns applied to agents produce products that feel unpredictable and untrustworthy. Users churn when they can't understand what the agent is doing.	Audit every agent feature against the Agent UX Checklist (Section 30.9). Fix all checklist failures before pilot launch.
Trust is architecture, not a feature. It is built through the totality of the experience — especially how failures are handled.	A single confident hallucination in a high-stakes moment can erase weeks of positive interactions. Failure handling deserves more design investment than the happy path.	Apply the Industrial Trust Design Framework (Section 31.8) to every agent feature. Calibration experience before first operational use is non-negotiable.
Automation bias is a real safety risk in industrial AI. Products must be designed to counteract it.	Operators under time pressure will follow AI recommendations without adequate verification unless the UX actively requires engagement with the evidence.	Build mandatory evidence engagement into high-stakes approval flows. Track override rate as a quality signal.
The Agent Builder pattern is one of the highest-leverage enterprise AI product patterns. It democratizes agent creation without democratizing risk.	Non-technical users who can build their own agents become power users and advocates. The platform multiplies its value with every agent created.	Prioritize Agent Builder V1 in the next product cycle. Start with a handful of high-quality templates users can deploy immediately.
The Autonomy Ratio is the most important long-term metric for agent product health.	An autonomy ratio that grows over time means the product is earning trust. Flat or declining means something is wrong with quality, UX, or trust design.	Add autonomy ratio to your weekly product review dashboard alongside task completion rate.
Industrial operators are a fundamentally different user population. Designing for desk workers produces products that fail in the field.	A product requiring deliberate reading will be ignored during high-cognitive-load moments when it matters most.	Conduct in-plant user research before finalizing mobile interface design. Watch an operator use the prototype during an active shift.

> **Up Next: Part VII — Production GenAI Metrics & Instrumentation**
>
> Part VII completes the measurement picture. It covers: the complete AI product metrics stack from infrastructure to business outcome, LLM evaluation frameworks in depth (RAGAS, G-EVAL, MT-Bench), building and maintaining evaluation datasets at scale, A/B testing methodology for AI products, the FinOps discipline for AI workloads (token economics, GPU cost modeling, cost-quality optimization), and the full AI product analytics stack.